Monetary Economics and Policy

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Chapter 2: Cash as a Medium of Exchange

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What We Will Learn

- Why cash (banknotes and coins) is introduced explicitly as a **medium of exchange** and how this differs from Chapter 1.
- How this affects money demand and price determination.
- Price-level determination under:
  - an interest-rate rule (FTPL and CBTPL),
  - control of monetary aggregates,
  - the **Gold Standard**,
  - **cryptocurrency competition**.
- What stablecoins are.

Outline

Currency as a Medium of Exchange

- This chapter introduces currency (cash and banknotes) explicitly as a **medium of exchange**.
- We will refer to currency as **money**, although money need not be currency (as discussed in Chapter 1).
- The understanding of money and its quantity has deep historical roots.
- **David Hume (1752)** theorized, in a metallic system, that:
  - An increase in the quantity of money is equivalent to an increase in the supply of gold and silver in a nation.
  - A higher money supply raises the domestic price level, reducing exports and increasing imports. As gold and silver flow out of the country, prices re-equalize internationally, restoring balance.

Historical Perspective on Money

- For most of history, money circulated in the form of **coins with precious metal content**, although primitive forms of money (e.g., wampum, tobacco) were used even into the colonial American period.

- A major medieval innovation was the ability to **economize on gold in transactions**, giving rise to instruments such as **bills of exchange**.

- Later, in Venice and Amsterdam, public banks issued **account money**—a form of non-physical money backed by metal reserves on their balance sheets and used as a means of payment.

- The **real bills doctrine**, developed by **Adam Smith (1776)**, argued that currency could be backed by **private short-term commercial claims** rather than only by gold—though the system as a whole still relied on metallic backing.

Intrinsically Worthless Currency I

- The emergence of an **intrinsically worthless paper-money system** represents a major discontinuity, even if historically it evolved gradually from fully backed to partially backed paper money.
- Under such a system, currency becomes a **self-referential claim**—a promise to be paid in units of itself, without intrinsic value or commodity backing.
- This contrasts sharply with earlier commodity-based systems in which money had intrinsic metallic value.

Intrinsically Worthless Currency II

- Besides central-bank-issued money, **private agents can also issue money** (e.g., bank deposits, stablecoins).
- Privately issued money typically represents a **claim on central bank money** or other high-quality liquid assets.
- Despite the absence of intrinsic value, money remains **essential in the payment system**, facilitating transactions and coordinating economic activity.

Monetarism: Historical Context

- The shift from commodity-backed to **intrinsically worthless fiat currency** paved the way for monetarism.
- Milton Friedman's *A Program for Monetary Stability* Friedman (1960) is widely viewed as the modern foundation of monetarist thought.
- After the collapse of the **Bretton Woods system** in the early 1970s, several central banks attempted to target money aggregates as the central focus of monetary policy.
- Canada, the United Kingdom, and the United States later abandoned strict monetary targeting due to instability in money–inflation relationships.

Monetarism: Ideas and Legacy

Milton Friedman (1960): *A Program for Monetary Stability*

- Modern foundation of monetarism.
- Monetarism holds that the **quantity of money strongly influences economic activity and the price level**.
- Monetary policy should therefore aim to **control the growth rate of the money supply**.
- Germany implemented a successful policy of targeting monetary aggregates for more than two decades, influencing the ECB's adoption of a monetary pillar in its strategy.

Model Overview

Economic Setting

- Infinite-horizon, closed economy with perfect foresight
- Single consumption good C with constant endowment Y

Agents

- Representative consumer
- Treasury
- Central Bank (CB)

One Currency = CB's liabilities

- Cash M_t
- Reserves X_t

Representative Consumer: Preferences

Intertemporal Utility

$$\sum_{t=t_0}^{\infty} \beta^{t-t_0} \left[U(C_t) + V\left(\frac{M_t}{P_t}\right) \right]. \tag{2.1}$$

Properties of $U(C_t)$

- $U(\cdot)$ concave and differentiable.
- Argument: consumption good C_t.
- $U_c(C_t) > 0, \quad U_{cc}(C_t) < 0.$

Properties of $V(M_t/P_t)$

- $V(\cdot)$ concave and differentiable.
- Argument: real money balances M_t/P_t.
- Satiation point $\bar{m} > 0$:

$$V_m\left(\frac{M_t}{P_t}\right) = 0 \quad \text{for } \frac{M_t}{P_t} \geq \bar{m}.$$

- $V_m(\cdot)$ is the derivative of $V(\cdot)$.

Budget Constraint

Budget Constraint at time t

$$\frac{B_t + X_t}{1 + i_t} + M_t + P_t C_t + T_t = B_{t-1} + X_{t-1} + M_{t-1} + P_t Y, \tag{2.2}$$

Where

- B_t: default-free one-period bond (unit face value), discounted at nominal rate i_t
- X_t: CB reserves, paying i_t^X
- M_t: cash (zero interest)
- P_t: price of consumption good
- T_t: lump-sum taxes (negative = transfers)
- Y: constant endowment of goods

Portfolio constraints

- Bonds: B_t can be held as asset (> 0) or issued as debt (< 0)
- Cash and reserves: $M_t \geq 0$, $X_t \geq 0 \; \forall \; t \geq t_0$

Borrowing Limit

- The consumer faces the borrowing limit:

$$-\frac{B_{t-1}}{P_t} \leq \frac{M_{t-1} + X_{t-1}}{P_t} + \sum_{j=0}^{\infty} R_{t,t+j} \left(Y - \frac{T_{t+j}}{P_{t+j}} \right) < \infty, \tag{2.3}$$

  where discount factor $R_{t,t+j}$ has the same definition as in Chapter 1.

- The consumer chooses $\{C_t, B_t, X_t, M_t\}_{t=t_0}^{\infty}$ with $C_t, X_t, M_t \geq 0$ to maximize intertemporal utility (2.1) subject to budget constraint (2.2) and the borrowing limit (2.3) at each time $t \geq t_0$ given initial conditions $B_{t_0-1}, X_{t_0-1}, M_{t_0-1}$.

Intertemporal Budget Constraint

- An equivalent representation of the consumer's problem can be obtained by replacing equation (2.3) with the intertemporal budget constraint:

$$\sum_{t=t_0}^{\infty} R_{t_0,t} \left(C_t + \underbrace{\frac{i_t}{1+i_t} \frac{M_t}{P_t}}_{\text{cost of holding money}} \right) \leq \frac{B_{t_0-1} + X_{t_0-1} + M_{t_0-1}}{P_{t_0}} + \sum_{t=t_0}^{\infty} R_{t_0,t} \left(Y - \frac{T_t}{P_t} \right). \quad (2.4)$$

- The present discounted value of consumption and the cost of holding money (the interest rate forgone on real money balances) must not exceed initial real wealth plus the present discounted value of net endowment.

Asset Limit Condition

- A third, equivalent formulation of the consumer's problem replaces constraint (2.4) with:

$$\lim_{t\to\infty}\left\{R_{t_0,t}\left(\frac{B_{t-1}+X_{t-1}+M_{t-1}}{P_t}\right)\right\}\geq 0. \tag{2.5}$$

- This condition requires that the limit of the discounted value of the household's asset holdings remains non-negative.

Consumer Optimality Conditions

- Define the Lagrange multiplier λ_t associated with constraint (2.2). Then $\lambda_t = U_c(C_t)/P_t$, where $U_c(\cdot)$ denotes the first derivative of the utility function $U(\cdot)$.
- The first-order condition (FOC) with respect to X_t or B_t is:

$$\frac{U_c(C_t)}{P_t} = \beta(1 + i_t)\frac{U_c(C_{t+1})}{P_{t+1}}. \tag{2.6}$$

- The FOC with respect to M_t implies:

$$\frac{U_c(C_t)}{P_t} = \frac{1}{P_t}V_m(\frac{M_t}{P_t}) + \beta\frac{U_c(C_{t+1})}{P_{t+1}}. \tag{2.7}$$

or, using equation (2.6):

$$1 = \frac{V_m(\frac{M_t}{P_t})}{U_c(C_t)} + \frac{1}{1 + i_t}. \tag{2.8}$$

Money Demand

- Using equation (2.8), we can express the demand for real money balances as:

$$\frac{M_t}{P_t} \geq V_m^{-1}\left(\frac{U_c(C_t)i_t}{1+i_t}\right) \equiv L\Bigg(\underset{(+)}{C_t},\ \underset{(-)}{i_t}\Bigg).$$

- When $i_t > 0$, the above relationship holds with equality. The demand for real money balances increases with consumption and decreases with the nominal interest rate.
- When $i_t = 0$, the marginal utility of real money balances $V_m(\cdot)$ is zero, implying that households may hold any arbitrary amount of real money balances above the satiation level.

Consumer Optimality: IBC or Transversality Condition

- In the optimal allocation the intertemporal budget constraint (2.4) holds with equality, which means that no resources are thrown away:

$$\sum_{t=t_0}^{\infty} R_{t_0,t}\left(C_t + \frac{i_t}{1+i_t}\frac{M_t}{P_t}\right) = \frac{B_{t_0-1} + X_{t_0-1} + M_{t_0-1}}{P_{t_0}} + \sum_{t=t_0}^{\infty} R_{t_0,t}\left(Y - \frac{T_t}{P_t}\right). \tag{2.9}$$

- Alternatively, the transversality condition holds, i.e., (2.5) holds with equality:

$$\lim_{t\to\infty}\left\{R_{t_0,t}\left(\frac{B_{t-1} + X_{t-1} + M_{t-1}}{P_t}\right)\right\} = 0. \tag{2.10}$$

Government

- The central bank's flow budget constraint is represented by:

$$\frac{B_t^C - X_t^C}{1 + i_t} - M_t^C = B_{t-1}^C - X_{t-1}^C - M_{t-1}^C - T_t^C, \tag{2.11}$$

where money M^C and reserves X^C are the central bank's liabilities and B^C the central bank's holdings of short-term default-free assets issued by the private sector or the treasury.

- The treasury's flow budget constraint is:

$$\frac{B_t^F}{1 + i_t} = B_{t-1}^F - T_t - T_t^C. \tag{2.12}$$

where B^F is the treasury's debt, T are lump-sum taxes and T^C are the nominal remittances received from the central bank.

Market Clearing

- In equilibrium, the markets for debt, reserves and money clear:

$$B_t^F = B_t^C + B_t, \tag{2.13}$$

$$X_t^C = X_t, \tag{2.14}$$

$$M_t^C = M_t, \tag{2.15}$$

for each $t \geq t_0$

- Equilibrium in asset markets together with the flow budget constraints of consumers and the government implies goods market equilibrium, $C_t = Y$ for each $t \geq t_0$.

Equilibrium Conditions

- The **central bank's budget constraint** is:

$$\frac{B_t^C - X_t}{1+i_t} - M_t = B_{t-1}^C - X_{t-1} - M_{t-1} - T_t^C. \tag{2.16}$$

- The **treasury's budget constraint** is:

$$\frac{B_t^F}{1+i_t} = B_{t-1}^F - T_t - T_t^C. \tag{2.17}$$

- The **bond market clearing condition** requires that:

$$B_t^F = B_t^C + B_t. \tag{2.18}$$

- The **intertemporal constraint on taxes**, identical to equation (1.12) in Chapter 1, is given by:

$$\frac{B_{t_0-1}^F}{P_{t_0}} = \sum_{t=t_0}^{\infty} \beta^{t-t_0} \left(\frac{T_t}{P_t} + \frac{T_t^C}{P_t} \right). \tag{2.19}$$

Equilibrium Conditions

- The **Fisher equation** (from the first-order condition (2.6) and goods market clearing) links the nominal interest rate to expected inflation:

$$1 + i_t = \frac{1}{\beta} \frac{P_{t+1}}{P_t}. \tag{2.20}$$

- The **money market equilibrium condition** (from the money demand equation) is:

$$\frac{M_t}{P_t} \geq L(Y_t, i_t), \tag{2.21}$$

  which holds with equality when $i_t > 0$.

- The **intertemporal resource constraint** (from combining equations (2.9) and (2.4)) ensures that the present value of real government resources equals the initial real value of its liabilities:

$$\sum_{t=t_0}^{\infty} \beta^{t-t_0} \left(\frac{T_t}{P_t} + \frac{i_t}{1+i_t} \frac{M_t}{P_t} \right) = \frac{B_{t_0-1} + X_{t_0-1} + M_{t_0-1}}{P_{t_0}}. \tag{2.22}$$

Equilibrium

An equilibrium is a set of sequences $\{P_t, i_t, X_t, M_t, B_t^C, T_t^C, B_t^F, T_t, B_t\}_{t=t_0}^{\infty}$ with $\{P_t, i_t, X_t, M_t, B_t^C\}_{t=t_0}^{\infty} \geq 0$ satisfying:

- (2.16), (2.17), (2.18), (2.20) and (2.21) holding at each $t \geq t_0$ and conditions (2.19) and (2.22) given initial values $B_{t_0-1}^C, B_{t_0-1}^F, X_{t_0-1}, M_{t_0-1}$ and that infinite sums in (2.19) and (2.22) are finite.
- $\{T_t\}_{t=t_0}^{\infty}$ satisfying (2.19) for any equilibrium sequences $\{P_t, T_t^C\}_{t=t_0}^{\infty}$ given initial condition $B_{t_0-1}^F$.

There are **four** degrees of freedom to specify government policy:

- We assume the treasury sets a path of taxes consistent with (2.17) and (2.19).
- The central bank simultaneously sets three of the four sequences $\{i_t, M_t, X_t, T_t^C\}_{t=t_0}^{\infty}$ but not simultaneously $\{i_t\}_{t=t_0}^{\infty}$ and $\{M_t\}_{t=t_0}^{\infty}$ for their dependence in (2.21).

What is Seigniorage?

- Seigniorage refers to the **profits earned by the issuer of currency**.
- When coins with metallic content were circulating, issuers such as sovereigns appropriated a small portion of the metal brought to the mint as payment for minting the coins and for affirming their value through the sovereign's portrait or coat of arms.
- In **modern monetary systems** with intrinsically worthless currency, seigniorage again represents the profit of the issuer.
- For a **central bank** that holds interest-bearing assets, profits arise when the return on assets exceeds the cost of its liabilities.
- Note that **Bitcoin** issues only liabilities and therefore does not generate profits.

Seigniorage in a Simple Model

- In the simple model of this chapter, the **central bank** holds assets B_t^C that earn an interest rate i_t.
- It issues currency in the form of **cash/banknotes** M_t bearing no interest, and **reserves** X_t bearing an interest rate i_t.
- The central bank earns **profits (seigniorage)** because it can finance, at zero cost (by issuing money), the purchase of default-free assets carrying a positive return.
- When the nominal interest rate is zero, these revenues are also zero.

Central Bank Net Worth

- The central bank's **net worth** at time t is defined as:

$$N_t^C = \frac{B_t^C - X_t}{1 + i_t} - M_t.$$

- The first term represents the present value of interest-bearing assets net of reserves, while the second term represents non–interest-bearing liabilities (cash).

- The **law of motion for net worth**, derived from the central bank's flow budget constraint (2.16), is:

$$N_t^C = N_{t-1}^C + \Psi_t^C - T_t^C,$$

where Ψ_t^C denotes central bank **profits (seigniorage)**.

- The term T_t^C represents remittances (or transfers) from the central bank to the treasury.

Central Bank Profits and Seigniorage

- Central bank profits are given by:

$$\Psi_t^C \equiv \frac{i_{t-1}}{1+i_{t-1}}(B_{t-1}^C - X_{t-1}) = i_{t-1}(N_{t-1}^C + M_{t-1}).$$

- Profits depend on the interest income received on assets minus the interest paid on reserves.
- Equivalently, seigniorage arises from the **non–interest-bearing liabilities** $(N_{t-1}^C + M_{t-1})$ multiplied by the nominal interest rate.
- When the nominal interest rate is zero, seigniorage is also zero.

Alternative Definition of Seigniorage

- The **literal definition of seigniorage** refers to the monetary resources that the central bank can extract from its operations.
- However, we can also define seigniorage as the **implicit cost for households** of holding certain securities at an interest rate below the market rate.
- These costs can be positive even when the central bank's accounting profits are zero.
- Formally, this alternative definition of seigniorage is represented by the infinite sum in equation (2.9):

$$\sum_{t=t_0}^{\infty} \beta^{t-t_0} \left(\frac{i_t}{1+i_t} \frac{M_t}{P_t} \right).$$

- This term corresponds to the **real resources households forgo** by holding money balances that do not earn the market rate of return.

Alternative Definition of Seigniorage

- This definition of seigniorage can be **positive even if the central bank makes no profit**, meaning that the literal definition has a value of zero.
- Consider the simple case in which the central bank **drops cash on the ground** without holding any assets. This can be Bitcoin creation.
- In this case:
$$M_t = M_{t-1} + T_t^C,$$
and profits are zero, but the infinite sum in equation (2.9) can still be positive.
- Intuitively, households incur a resource cost for holding money that bears no interest.
- *Note:* In this example, net worth satisfies $N_t^C = -M_t$.

Interest-Rate Policy and FTPL

- Assume that the central bank sets policy in terms of $\{i_t, X_t\}_{t=t_0}^{\infty}$, so that the path of money supply becomes endogenous. The interest rate policy is the same as in Chapter 1:

$$1 + i_t = \max\left\{\frac{1}{\beta}\left(\frac{P_t}{\bar{P}}\right)^{\phi}, 1\right\}, \qquad \phi \geq 0,\ \bar{P} > 0.$$

- Assume further that the central bank **fully backs the treasury's debt** (Fiscal Theory of the Price Level, FTPL).

- Modify the **tax policy** to be constant, taking into account the transfer of seigniorage revenues obtained from issuing money:

$$\frac{T_t}{P_t} = (1-\beta)\tau - i_{t-1}\frac{M_{t-1}}{P_t}, \qquad (\tau > 0). \tag{2.23}$$

Interest-Rate Policy and FTPL

- Inserting the tax policy (2.23) into equation (2.22) yields:

$$\frac{B_{t_0-1} + X_{t_0-1}}{P_{t_0}} + (1 + i_{t_0-1})\frac{M_{t_0-1}}{P_{t_0}} = \tau.$$

- This determines the **initial price level** P_{t_0}, given the initial conditions B_{t_0-1}, X_{t_0-1}, M_{t_0-1}, and i_{t_0-1}.
- The full sequence of prices follows from the combination of the **interest-rate policy** and the **Fisher equation**.
- The value of τ determines whether the equilibrium is **inflationary**, **deflationary**, or one with a constant price level $\bar{P}$.
- The sequence of money supply is determined by equation (2.21).
- Despite the presence of cash in the economy, **price determination under an interest-rate policy remains unchanged** in its fundamental mechanism with respect to that of Chapter 1.

Interest-Rate Policy and CBTPL

- Assume that the treasury's debt is **not backed by the central bank** (CBTPL).
- The interest-rate policy is the same as before, while the **remittances policy** is given by:

$$\frac{T_t^C}{P_t} = (1-\beta)\tau^C + i_{t-1}\frac{M_{t-1}}{P_t}, \tag{2.24}$$

- Prices are determined by the same mechanism as in Chapter 1. The **initial price level** is given by:

$$P_{t_0} = \frac{B_{t_0-1}^C - X_{t_0-1} - (1+i_{t_0-1})M_{t_0-1}}{\tau^C} = \frac{N_{t_0-1}^C}{\tau^C(1+i_{t_0-1})}.$$

- Owing to seigniorage revenues, the parameter τ^C can take **negative values** with a negative net worth, while remittances $\frac{T_t^C}{P_t}$ remain positive.

Seigniorage as a Real Backing of Currency

- The term $\frac{i_t}{1+i_t}\frac{M_t}{P_t}$ represents a **real revenue** for the central bank, mirroring the real resources that consumers forgo by holding money balances.
- This revenue can be used to retrieve resources that **back the value of currency**.

Example:

- Consider a nominal remittances policy $T_t^C = (1-\beta)T^C \geq 0$ under a pure interest-rate peg $1 + i_t = 1/\beta$ (previously unable to determine the price level).
- Substituting $T_t^C = (1-\beta)T^C$ into (2.22) using (2.19) , together with money-market equilibrium (2.21) and the constant price level implied by the interest-rate policy, determines the intial price level:

$$P_{t_0} = \frac{T^C + (X_{t_0-1} + M_{t_0-1} - B^C_{t_0-1})}{L\left(Y, \frac{1}{\beta} - 1\right)}. \tag{2.25}$$

Seigniorage as a Real Backing of Currency

- With seigniorage, a positive P_{t_0} can be achieved even with a negative net worth.
- Seigniorage resources can provide real backing to the value of currency even with no asset holdings and zero transfers.
- The key feature for seigniorage to establish a unique price level is the **essential role of currency as a means of payment**, otherwise a non-monetary equilibrium ($P_t = \infty$) can still exist.
- One way to rule out the non-monetary equilibrium is to assume:

$$\lim_{P_t \to \infty} \frac{1}{P_t} V_m\left(\frac{M_t}{P_t}\right) > 0. \tag{2.26}$$

- This assumption requires that cash remains **essential even at very high interest rates**.

Note on Seigniorage

- Consider central bank's net worth:

$$N_t^C \equiv \frac{B_t^C - X_t}{1 + i_t} - M_t,$$

and its law of motion in real terms:

$$\frac{N_t^C}{P_t} = (1 + i_{t-1})\frac{N_{t-1}^C}{P_t} + i_{t-1}\frac{M_{t-1}}{P_t} - \frac{T_t^C}{P_t}.$$

- Iterating it forward and using the equilibrium value for $R_{t,t_0} = \beta^{t-t_0}$ we obtain:

$$\frac{X_{t_0-1} + M_{t_0-1}}{P_{t_0}} + \sum_{t=t_0}^{\infty}\beta^{t-t_0}\left(\frac{T_t^C}{P_t}\right) = \frac{B_{t_0-1}^C}{P_{t_0}} + \sum_{t=t_0}^{\infty}\beta^{t-t_0}\left(\frac{i_t}{1+i_t}\frac{M_t}{P_t}\right). \tag{2.27}$$

Note on Seigniorage

- Equation (2.27) is not a solvency condition for the central bank but rather an equilibrium condition.
- The real value of the central bank's liabilities, in combination with the present-discounted value of real remittances (left-hand side), must **in equilibrium** be equal to the sum of asset holdings and seigniorage (right-hand side).
- Note that the seigniorage term can be positive even if the central bank does not make any profit nor hold any asset.

Targeting Money Aggregates

- Consider a different policy targeting money aggregates and a simple one of constant money supply, $M_t = M \; \forall t \geq t_0$.
- Using $C_t = Y$ inside the first order condition (2.7) implies:

$$m_{t+1} = \frac{m_t}{\beta}\left(1 - \frac{V_m(m_t)}{U_c(Y)}\right), \tag{2.28}$$

  with $m_t = \frac{M}{P_t}$.
- Equation (2.28) is a first-order non-linear difference equation in real money balances that has infinite solutions, depending on the initial real money balances m_{t_0} or price level P_{t_0}, which is not determined.
- Even with a constant money supply, and in general with a policy controlling money aggregates, there are many paths for the price level consistent with (2.28).
- $\Rightarrow$ other policy tools are needed to uniquely determine the price level as in the case of interest rate policies.

Money Aggregates and Additional Policy Tools

- If the central bank backs the treasury (FTPL), (2.22) is an equilibrium condition and the treasury could set a tax policy like (2.23) to determine the initial price level.
- If the central bank does not back the treasury(2.27) is an equilibrium condition and the central bank can set an appropriate remittances policy to determine the initial price level.
- An alternative mechanism for backing currency is to threat redemption with a commodity (gold).

Friedman Rule

- What is the **optimal monetary policy** in a model where households derive utility from real money balances?
- Holding real money balances implies a **real cost** for households due to foregone earnings in the bond market.
- Since producing cash is costless, a **benevolent policymaker**, maximizing household welfare, would want to equalize the marginal benefit of money $V_m(\cdot)$ with its zero marginal cost. This can be achieved by **satiating the economy** with real money balances, i.e.
$$\frac{M_t}{P_t} \geq \bar{m},$$
which also reduces the consumer's cost of holding money to zero, with $i_t = 0$.
- This is the famous **Friedman rule**: to implement $i_t = 0$, the policymaker should target a **negative money growth rate** and achieve a **rate of deflation** equal to the rate of time preference,
$$\frac{P_{t+1}}{P_t} = \beta.$$

Gold Standard

- Under the gold standard, currency is backed by gold holdings: the central bank commits to fixing the price of gold and allowing the free conversion of currency into gold at that price.

- Consider a generalization of household preferences as in Chapter 1:

$$\sum_{t=t_0}^{\infty} \beta^{t-t_0} \left[U(C_t) + L\left(\frac{M_t}{P_t}\right) + Z(g_t) \right]$$

  where $Z(g_t)$ is the utility the household derives from holding gold g_t.

- The household's budget constraint becomes:

$$\begin{aligned} &\frac{B_t}{1+i_t} + M_t + P_{g,t} g_t + P_t C_t + T_t \\ &\quad = B_{t-1} + M_{t-1} + P_{g,t} g_{t-1} + P_t Y_t + P_{g,t}(g_t^S - g_{t-1}^S), \end{aligned}$$

  where P_g is the gold price and g^s is the stock of gold in the economy.

Gold Standard: First-Order Condition for Gold

- The household FOCs (2.6) and (2.7) are augmented by the FOC with respect to gold holdings:

$$\frac{P_{g,t}}{P_t} = \frac{Z_g(g_t)}{U_c(C_t)} + \frac{1}{1+r_t}\frac{P_{g,t+1}}{P_{t+1}}. \tag{2.28}$$

- This FOC links today's relative price of gold to its marginal utility and its discounted future relative price.

Gold Standard: Mechanism

- Under the gold standard, the monetary authority fixes the price of gold and is ready to buy or sell gold at that price, let's say:

$$P_{g,t} = 1.$$

- Money is fully backed by central bank gold holdings:

$$M_t = P_{g,t} g_t^C = g_t^C.$$

- With goods market equilibrium ($C_t = Y$) and gold market equilibrium ($g_t^S = g_t + g_t^C$), the FOCs (2.7) and (2.28) become:

$$\frac{1}{P_t} = \frac{1}{P_t} \frac{L_m\left(\frac{g_t^C}{P_t}\right)}{U_c(Y)} + \frac{\beta}{P_{t+1}}, \tag{2.29}$$

$$\frac{1}{P_t} = \frac{Z_g(g_t^S - g_t^C)}{U_c(Y)} + \frac{\beta}{P_{t+1}}. \tag{2.29}$$

Determination of the Price Level

- Equating the two expressions for $\frac{1}{P_t}$ yields:

$$P_t = \frac{L_m\left(\frac{g_t^C}{P_t}\right)}{Z_g(g_t^S - g_t^C)}. \tag{2.30}$$

- Price movements are **constrained** as long as the supply of gold is bounded.
- By iterating (2.29) forward, one obtains P_t and g_t^C at each date, jointly with (2.30).
- The value of money is anchored by:
  - the marginal utility of gold to households, and
  - the limited supply of gold, which constrains the price level.

Empirical Relevance of the Gold Standard

- The model's implications align with the observed evidence of **price stability** under the Classical Gold Standard (1880–1914).
- However, this stability came with a cost: the price level was highly sensitive to both monetary and real shocks, making inflation more volatile than under inflation targeting.
- Historically, gold discoveries (e.g., California 1848) caused persistent price level increases, consistent with the model.

Cryptocurrency Competition: Consumer

- We extend the model to include **privately issued money** M_t^* in addition to government money M_t.

Household utility:

$$\sum_{t=t_0}^{\infty} \beta^{t-t_0} \left[U(C_t) + V\left(\frac{M_t}{P_t} + \frac{M_t^*}{P_t^*} \right) \right].$$

Budget constraint:

$$\frac{B_t}{P_t(1+i_t)} + \frac{M_t}{P_t} + \frac{M_t^*}{P_t^*} + C_t \leq \frac{B_{t-1}}{P_t} + \frac{M_{t-1}}{P_t} + \frac{M_{t-1}^*}{P_t^*} + Y - \frac{T_t}{P_t} + \frac{Tr_t^*}{P_t^*}. \tag{2.30}$$

Tr_t^* are transfers from the private issuer in its own currency.

First-Order Conditions: M_t and M_t^*

- At each time $t \geq t_0$, the first-order conditions with respect to government money M_t and private money M_t^* are:

$$\frac{U_c(C_t)}{P_t} = \frac{1}{P_t} V_m\left(\frac{M_t}{P_t} + \frac{M_t^*}{P_t^*}\right) + \beta \frac{U_c(C_{t+1})}{P_{t+1}} + \psi_t, \tag{2.31}$$

$$\frac{U_c(C_t)}{P_t^*} = \frac{1}{P_t^*} V_m\left(\frac{M_t}{P_t} + \frac{M_t^*}{P_t^*}\right) + \beta \frac{U_c(C_{t+1})}{P_{t+1}^*} + \psi_t^*. \tag{2.32}$$

- $\psi_t \geq 0$ and $\psi_t^* \geq 0$ are multipliers for $M_t \geq 0$ and $M_t^* \geq 0$.

Kuhn–Tucker Conditions and Currency Dominance

- The Kuhn–Tucker conditions are:

$$\psi_t M_t = 0, \qquad \psi_t^* M_t^* = 0, \qquad \psi_t, \psi_t^* \geq 0.$$

- Combining (2.31) and (2.32) yields:

$$\psi_t P_t - \psi_t^* P_t^* = \beta U_c(C_{t+1}) \left(\frac{P_t^*}{P_{t+1}^*} - \frac{P_t}{P_{t+1}} \right).$$

- If private-currency inflation is higher, $\frac{P_{t+1}^*}{P_t^*} > \frac{P_{t+1}}{P_t}$, then the RHS is negative, implying $\psi_t^* > 0$ and therefore $M_t^* = 0$.
- Conversely, if private-currency inflation is lower, government money is not used.
- If inflation rates are equal, both currencies may coexist.

Equilibrium

- At the optimum, and in equilibrium, the intertemporal budget constraint of the consumer holds, implying, by using market clearing equations $C_t = Y$, $B_t = B_t^g$, $M_t = M_t^g$ and $M_t^* = M_t^{*p}$ for each $t \geq t_0$, that:

$$\frac{M_{t_0-1} + B_{t_0-1}}{P_{t_0}} + \frac{M^*_{t_0-1}}{P^*_{t_0}} = \sum_{t=t_0}^{\infty} \beta^{t-t_0} \left[\frac{i_t}{1+i_t} \frac{M_t}{P_t} + \frac{i^*_t}{1+i^*_t} \frac{M^*_t}{P^*_t} + \frac{T_t}{P_t} - \frac{Tr^*_t}{P^*_t} \right]. \tag{2.33}$$

- Equation (2.33) is the intertemporal resource constraint of the economy. Note that it does not anymore resemble an intertemporal budget constraint for the government!

Equilibrium

- Monetary policy for the private issuer is specified by a constant growth rate of private money:

$$M_t^* = (1 + \vartheta^*)\, M_{t-1}^*, \qquad t \geq t_0.$$

- The government has an additional degree of freedom, as it also issues **interest-bearing securities**.

- We assume the government sets a constant interest rate $i_t = i$ and follows the tax policy:

$$\frac{T_t}{P_t} = (1 - \beta)\tau - i\, \frac{M_{t-1}}{P_t}, \quad \text{with } \tau > 0. \tag{2.38}$$

- This interest-rate policy implies **constant inflation** through the Fisher equation:

$$\Pi = \beta(1 + i),$$

and hence the price level evolves as:

$$P_{t+1} = \Pi P_t, \qquad t \geq t_0.$$

Equilibrium

To determine the initial price level P_{t_0} and the sequences $\left\{P_t^*, M_t, B_t\right\}_{t_0}^{\infty}$ the following equilibrium conditions are relevant:

$$\frac{1}{P_t}\frac{i}{1+i} = \frac{1}{P_t}V_m\left(\frac{M_t}{P_t}+\frac{M_t^*}{P_t^*}\right)+\psi_t \tag{2.34}$$

$$\frac{1}{P_t^*} = \frac{1}{P_t^*}V_m\left(\frac{M_t}{P_t}+\frac{M_t^*}{P_t^*}\right)+\frac{\beta}{P_{t+1}^*} \tag{2.35}$$

$$\beta\frac{(1+i)M_t+B_t}{P_{t+1}} = \frac{(1+i)M_{t-1}+B_{t-1}}{P_t}-(1-\beta)\tau \tag{2.36}$$

$$\frac{(1+i)M_{t_0-1}+B_{t_0-1}}{P_{t_0}}+\frac{M_{t_0-1}^*}{P_{t_0}^*} = \tau+\sum_{t=t_0}^{\infty}\beta^{t-t_0}\left[\frac{i_t^*}{1+i_t^*}\frac{M_t^*}{P_t^*}-\vartheta^*\frac{M_{t-1}^*}{P_t^*}\right]. \tag{2.37}$$

together with $\psi_t, P_t, P_t^*, M_t \geq 0$ and Kuhn-Tucker condition $\psi_t M_t = 0$.

Equilibrium with Only Government Money is Used

- There is always an equilibrium in which only government money is used.
- First-order condition (2.35) is consistent with an infinite price level for private currency.
- Real taxation with interest-rate policy are always able to give a positive value to government currency.
- The inflation rate, $\Pi_t = P_t/P_{t-1}$, is set at $\Pi = \beta(1+i)$ and the price level is determined using (2.37) to obtain

$$\frac{(1+i)M_{t_0-1} + B_{t_0-1}}{P_{t_0}} = \tau. \tag{2.38}$$

- This result is a consequence of the asymmetries in the currencies. What limits the private currency is the absence of resources that can back it.

Equilibrium with Both Currencies Used

- When both currencies are used $\Pi = \Pi^*$.
- For $i > 0$, (2.34) implies

$$V_m\left(\frac{M_t}{P_t} + \frac{M_t^*}{P_t^*}\right) = \frac{i}{1+i}. \tag{2.39}$$

- It follows that the sum of real money balances should always be equal to a generic constant $c > 0$:

$$\frac{M_t}{P_t} + \left(\frac{1+\vartheta^*}{\Pi}\right)^{t+1-t_0} \frac{M_{t_0-1}^*}{\Pi P_{t_0}^*} = c. \tag{2.40}$$

- For an equilibrium with two currencies to exist, it must be that $1 + \vartheta^* \leq \beta(1+i) = \Pi$ otherwise government money supply will be negative within some finite period of time.

Gresham's Law

- When $1 + \vartheta^* < \Pi$, real money balances in private currency shrink over time and converge to zero in the long run, while real balances in government currency rise to the upper bound c.
- The growth rate of government money is higher than the inflation rate Π and therefore higher than that of private money.
- This represents an example of *Gresham's law*, where the "bad" money (with the higher growth rate) crowds out the "good" money.

Both Currencies are Used

- When $i = 0$ and $\Pi = \Pi^* = \beta$ there is no bound on real money balances.
- An additional equilibrium requirement is that the following term must be finite:

$$\sum_{t=t_0}^{\infty} \beta^{t-t_0} \left(\frac{Tr_t^*}{P_t^*} \right) = \vartheta^* \sum_{t=t_0}^{\infty} \beta^{t-t_0} \left(\frac{M_{t-1}^*}{P_t^*} \right) = -\vartheta^* \frac{M_{t_0-1}^*}{P_{t_0}^*} \sum_{t=t_0}^{\infty} (1 + \vartheta^*)^{t-t_0}. \tag{2.41}$$

- This requires $\vartheta^* = 0$.
- When $i = 0$ and $\vartheta^* = 0$, **private money becomes a bubble** whose real value grows over time at the rate $1/\beta$.

Exchange Rate Indeterminacy

- When both currencies are used, the exchange rate $\mathcal{E}_t$ is **indeterminate** and remains constant at a level not pinned down in equilibrium.
- This arises because equal inflation rates, $\Pi = \Pi^*$, imply a fixed nominal exchange rate $\mathcal{E}_t = \mathcal{E}$ without determining its level.
- When $i = 0$ and $\vartheta^* = 0$, exchange rate indeterminacy **causes** price-level indeterminacy.

Only Private Money is Used

- Equilibria can exist where **only private money** is used as a medium of exchange, though **note** that there is no equilibrium where government currency is worthless, i.e. in equilibrium it is always the case that $0 < P_t < \infty$ at each point in time.
- In this equilibrium, inflation on private money must be lower than that on government money: $\frac{P^*_{t+1}}{P^*_t} < \frac{P_{t+1}}{P_t}$.
- When $M_t = 0$, equation (2.35) becomes

$$m^*_{t+1} = \frac{1+\vartheta^*}{\beta}(1 - V_m(m^*_t))m^*_t, \tag{2.42}$$

where $m^*_t \equiv M^*_t / P^*_t$.

Only Private Money is Used

- The non-linear difference equation (2.42) admits multiple equilibria.
- In the stationary equilibrium $\tilde{m}^*$ satisfies: $1 - V_m(\tilde{m}^*) = \frac{\beta}{1+\vartheta^*}$, which requires $1 + \vartheta^* < \Pi$.
- In this equilibrium, $\Pi^* = 1 + \vartheta^*$ and $\Pi > \Pi^*$, therefore private money can exclude government money as a medium of exchange.
- There can also be equilibria with **rising real money balances**, provided $1 + \vartheta^* < \Pi$. In this case, Π_t^* decreases over time until real balances reach full satiation.

Cryptocurrency Competition: Main Conclusion

- Issuing private money as a **"strong" currency**, with money growth lower than inflation in government currency,

$$1 + \vartheta^* < \Pi,$$

  creates the possibility that government currency is **excluded** as a **medium of exchange**.

- However, $1 + \vartheta^* < \Pi$ is **not sufficient** for exclusive private-currency usage.

- We showed additional equilibria in which:
  - private money is **worthless** (not used), or
  - private and government money **coexist**, with private money losing purchasing power over time.

Cryptocurrency Competition: Main Conclusion

- The results can also be read in reverse: **how can the government crowd out private money** as a medium of exchange?

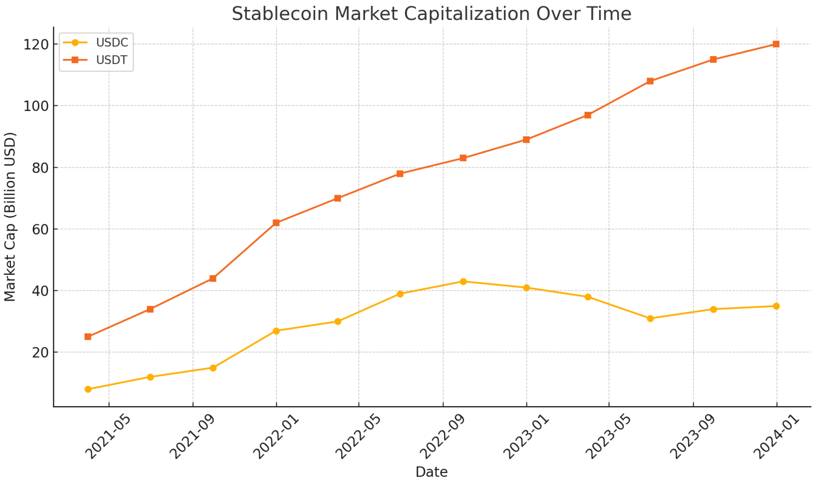
- The government must keep inflation **low**, so that government currency remains the more attractive medium of exchange:

$$\Pi < 1 + \vartheta^*.$$

- Hence, **currency competition** can discipline monetary policy:
  - it helps keep inflation low,
  - it reduces seigniorage revenues (monopoly rents),
  - it pushes the economy closer to the **efficient allocation** (Friedman rule) when multiple issuers compete in the liquidity market.

Stablecoins

- Stablecoins such as **USD Tether** and **USD Circle** are promises of **synthetic dollars** — they aim to replicate the value of the U.S. dollar.
- They achieve this by investing in **liquid assets**, typically short-term U.S. Treasury bills (maturity below 90 days) or other high-quality liquid assets. (The U.S. *Genius Act* enforces these requirements.)
- As a result, their exchange rate relative to the dollar stays **very close to 1**, which is the intended stable parity.
- Replicating the dollar through investment in interest-bearing securities generates **profits for stablecoin issuers**.
- Rebating these profits to users could, in principle, **crowd out government currency as a medium of exchange**.


Stablecoin Market Capitalization Over Time
USDC
USDT
Market Cap (Billion USD)
120
100
80
60
40
20
2021-05
2021-09
2022-01
2022-05
2022-09
2023-01
2023-05
2023-09
2024-01
Date

Stablecoin Reserve Comparison (2024 Snapshot)

Metric	**USDC (Circle)**	**USDT (Tether)**
Market Cap	$34.7B	$118–163B
Liquidity Ratio	~93% liquid	~81.5% liquid
Main Holdings	Treasuries, repo	Treasuries, crypto, gold
Audit Frequency	Monthly attestations & audits	Monthly attestations only
Stress Events	Mar 2023 (SVB exposure)	Oct 2018, 2021, Terra 2022

Source: Circle, Tether transparency reports, IMF Crypto Monitor (2025).

Launching a Stablecoin

- A private consortium issues a currency fully **backed by a basket of risk-free securities** denominated in government currency.
- The consortium buys and sells the private currency at an arbitrarily chosen exchange rate $\mathcal{E}_t$, investing proceeds $M_t^* \mathcal{E}_t$ in safe bonds. These bonds yield $(1 + i_t) M_t^* \mathcal{E}_t$, from which a management fee $\phi_t^f M_t^* \mathcal{E}_t$ is deducted.
- To credibly promise repurchases of its currency, the exchange rate must satisfy:

$$\mathcal{E}_{t+1} = (1 + i_t - \phi_t^f)\, \mathcal{E}_t. \tag{2.50}$$

- **Implications:**
  - If $\phi_t^f < i_t$: $\mathcal{E}_t$ **appreciates**; government money is crowded out.
  - If $\phi_t^f > i_t$: only government currency is used.
  - If $\phi_t^f = i_t$: both currencies coexist — the private currency is a **stablecoin**.

Stablecoins: Key Takeaways

- A stablecoin is a **privately issued synthetic dollar** designed to maintain a near-par exchange rate with government currency ($\mathcal{E}_t \approx 1$).
- Credible stability requires **high-quality backing** (e.g., short-term Treasuries) and a **redemption commitment** at the chosen exchange rate.
- Backing generates a **carry return** for the issuer. Under the pricing condition

$$\mathcal{E}_{t+1} = (1 + i_t - \phi_t^f)\mathcal{E}_t,$$

  the management fee ϕ_t^f determines whether the private currency appreciates, is stable, or depreciates.
- If issuers rebate part of the backing returns to users (low ϕ_t^f), stablecoins can become a **high-return medium of exchange** and potentially **crowd out government money**.
- Conversely, maintaining government currency dominance requires keeping inflation low and/or regulating stablecoin backing and payout policies.

Main Takeaways

- A fiduciary-currency system is a historical discontinuity: currency becomes a **self-referential claim**, yet remains essential for payments.
- Controlling monetary aggregates alone cannot uniquely determine the price level — capital backing, taxes, or convertibility must complete the system.
- Seigniorage may also provide a **real backing** for currency.
- Under the Gold Standard, the price level is anchored by the **value of gold** and by its limited supply.
- The **Friedman rule** prescribes satiating real money balances and achieving deflation at rate β.
- Currency competition shows that a “strong” private currency (or a stablecoin with backing) can **crowd out government money**, disciplining inflation.

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